Password Strength Checker using Python

Platform: Windows

Language: Python

Library: re (Regular Expression library)

1. Project Title

Password Strength Checker using Python

2. Objective

The main objective of this project is to develop a Password Strength Checker that evaluates the strength of a user's password based on certain criteria such as length, use of uppercase and lowercase letters, numbers, and special characters. This project helps in understanding basic Python programming and security concepts.

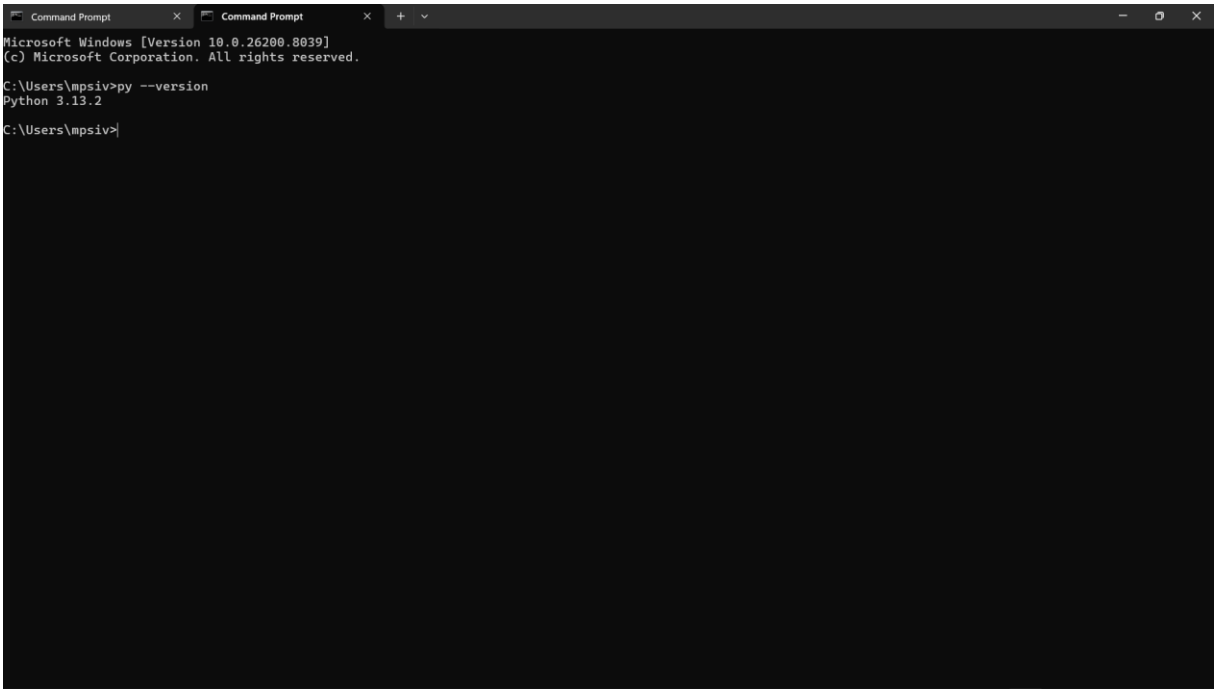
3. Software Requirements

The following software requirements are needed for the project:

- Windows operating system
- Python 3.x
- Command Prompt / Terminal

4. Step 1 – Check Python installation

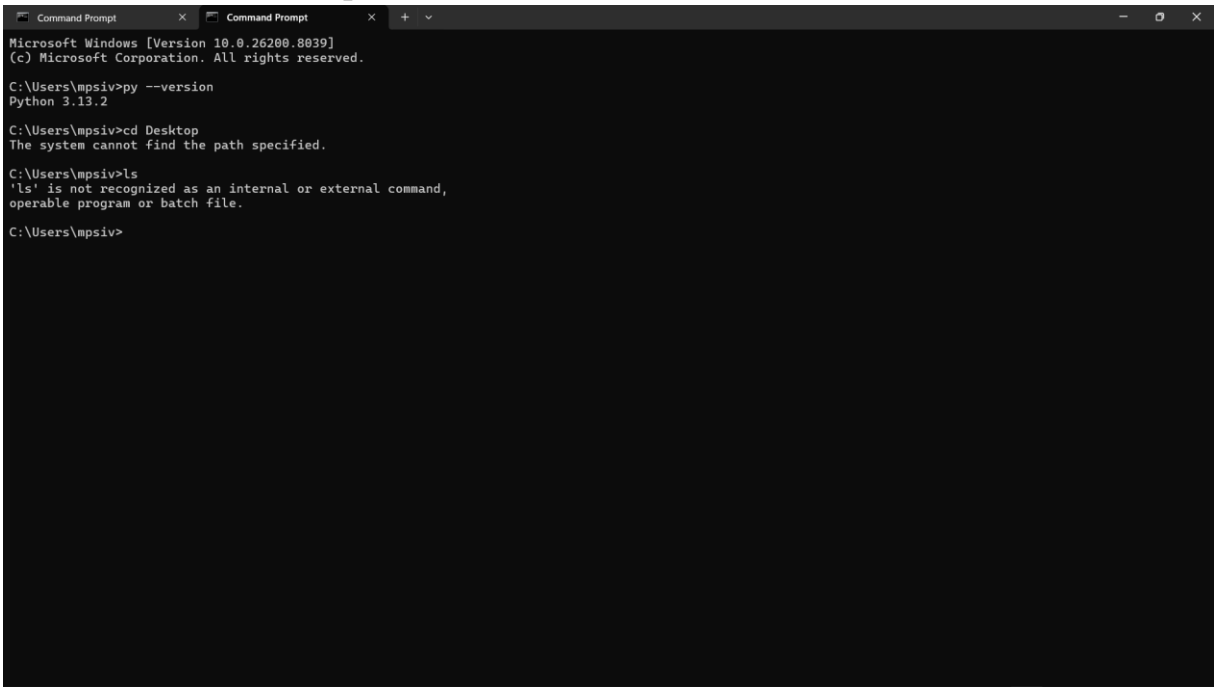
Command: py --version



This image shows that Python is installed successfully in the Windows system. The py --version command is used to verify the installed version of Python before starting the project.

5. Step 2 – Move to Desktop

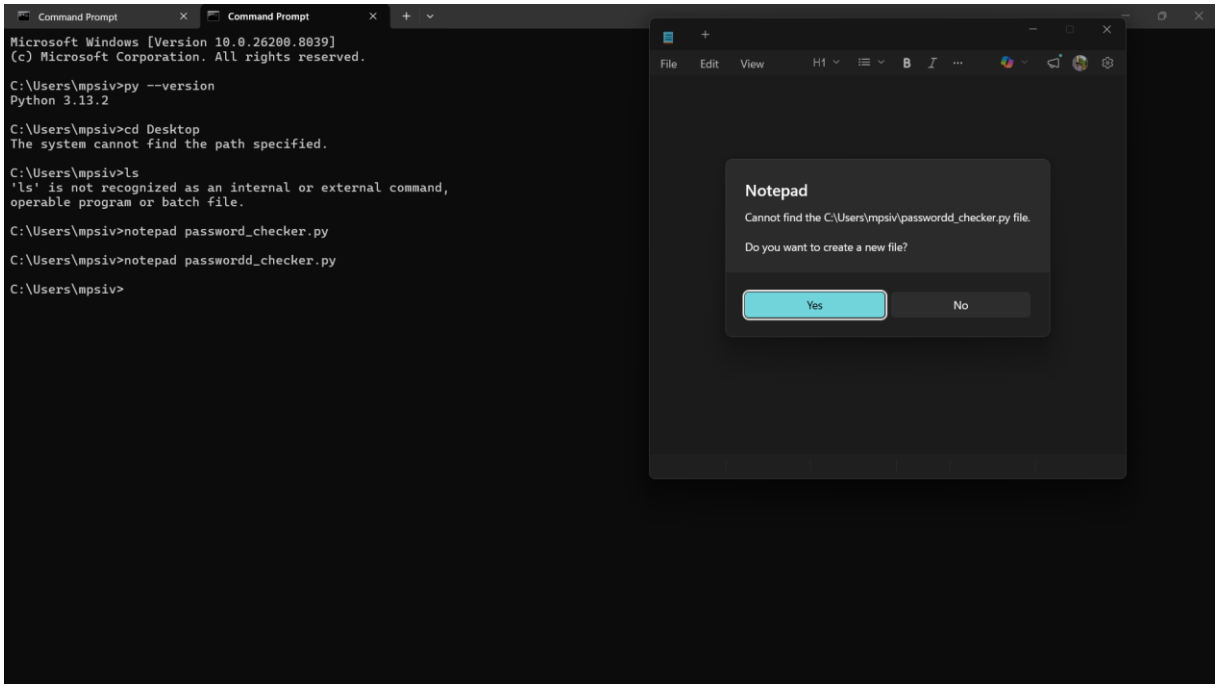
Command: cd Desktop



This image shows that the Command Prompt is moved to the Desktop folder. The Desktop location is used to create and run the project file easily.

6. Step 3 – Create the Python file

Command: notepad password_checker.py

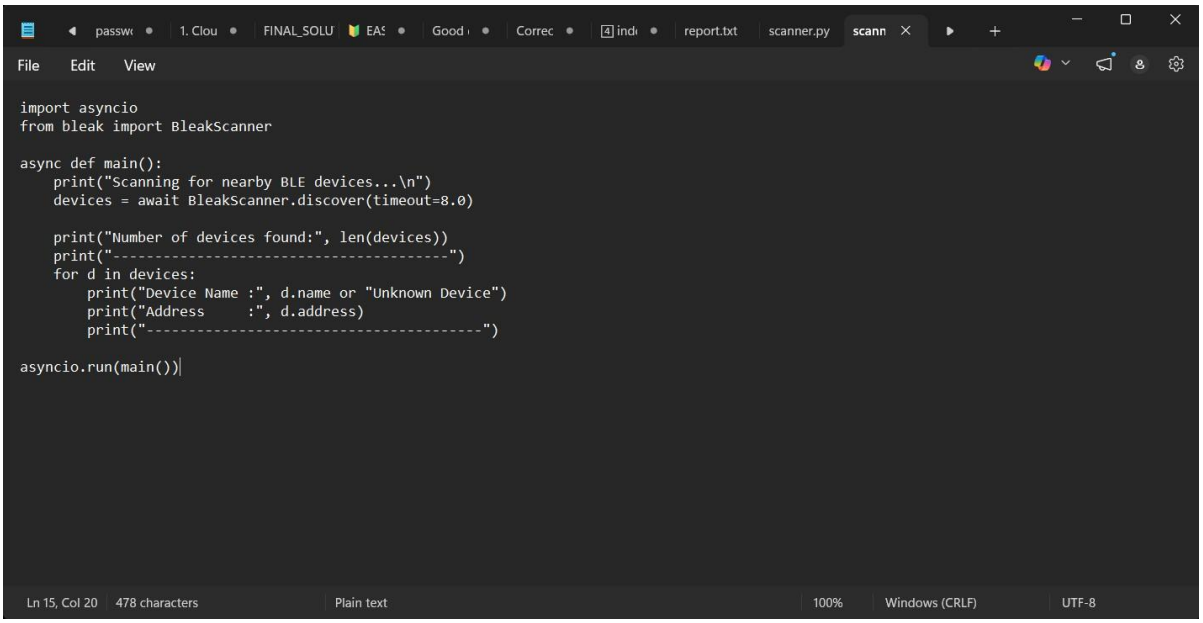


This screenshot shows the creation of the project source file scanner.py using Notepad. The file password_checker.py will contain the program code.

7. Step 4 – Write the Python code

scanner.py:

```
import re

password = input("Enter your password: ")

strength = "Weak"

if (len(password) >= 8 and
    re.search("[a-z]", password) and
    re.search("[A-Z]", password) and
    re.search("[0-9]", password) and
    re.search("[@#$%^&+=!]", password)):
    strength = "Strong"

elif len(password) >= 6:
    strength = "Medium"

print("Password Strength:", strength)
```



This image shows the Python code written for the Password Strength Checker project. The code checks the password strength based on length and character conditions.

8. Step 5 – Save the file

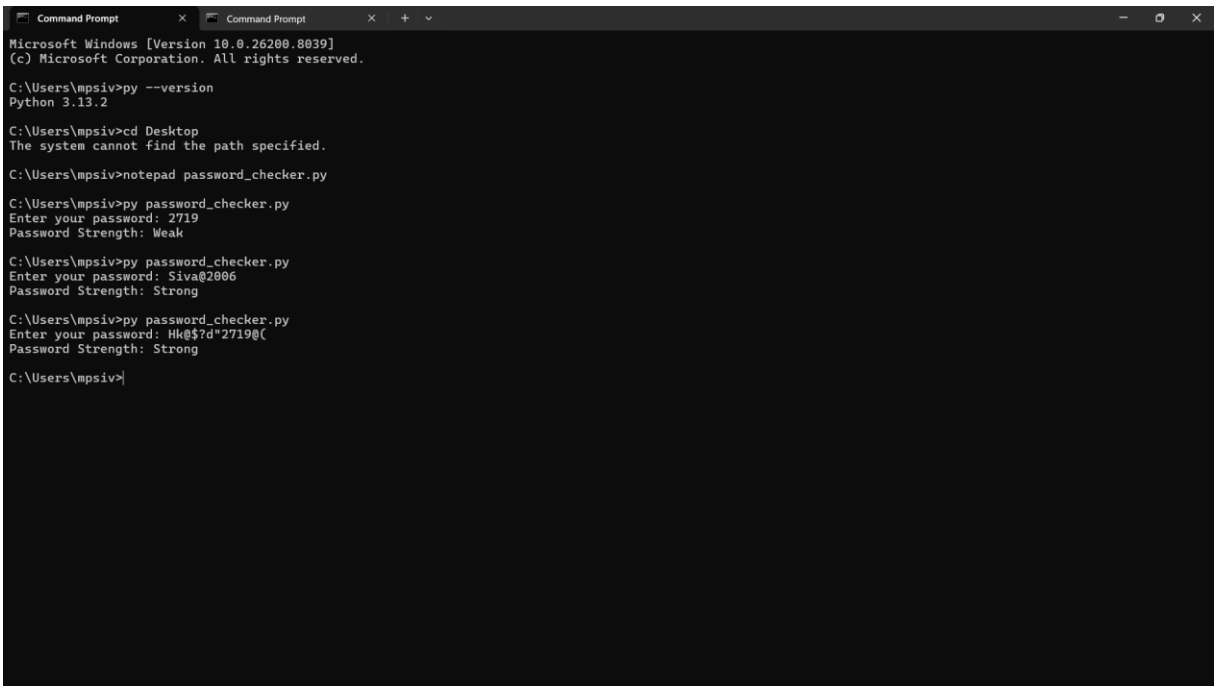
Action

Press: Ctrl + S

Then close Notepad.

9. Step 6 – Run the Python file

Command: py password_checker.py



This image shows the execution of the Python program using Command Prompt. The program starts and asks the user to enter a password.

10. Step 7 – Verify project success

The successful output confirms that the Password Strength Checker project is working correctly in the Windows environment. The program was able to accept user input, analyze the password based on defined criteria, and display the strength of the password as Weak, Medium, or Strong.